Agent-Population Scaling and Decentralized Team Recruitment in Alem

Combined E0–E3 Methods, Results, and Preregistered Evaluation Report

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Abstract

This report consolidates a staged investigation of physical-agent scaling, decentralized team recruitment, and team-scoped communication in Alem. The completed evidence is limited to two provider-free engineering studies. E0 exercised the canonical evaluator at $N \in \{1, 2, 3, 4, 6\}$ for two seeds and ten ticks and passed every structural, accounting, replay, and renderer gate. E0.1 requested 200 ticks with the same scripted `Noop` policy. All structural gates again passed, but only seven of ten episodes reached the requested horizon; three ended normally after all agents died. E0.1 is therefore a structural **PASS** and a literal full-horizon **FAIL**. Neither study measures useful behavior.

The remaining studies are preregistered and have no results at the time of this report. E1 will measure the unchanged GPT-5.4 nano Source baseline as physical population changes. E2 will isolate decentralized recruitment in a deterministic six-agent microbenchmark with bounded public control messages and team-private conversation. E3 will transfer the promoted mechanism to Alem behind an opt-in topology while preserving Source behavior. The report fixes hypotheses, paired seeds, outcomes, promotion rules, audit requirements, and implementation risks before those results are observed. In particular, population scaling is treated as a fixed-world descriptive curve rather than a pure causal agent-count effect, and no claim about decentralized recruitment will be made unless routing, agreement, validity, and replay gates pass.

1 Program status and claim boundary

Stage	Status	Current evidence or blocker
E0: 10-tick compatibility	**PASS**	Ten provider-free episodes completed; all structural gates and five renderer smokes passed.
E0.1: 200-tick extension	**PASS** structural / **FAIL** horizon	Ten complete artifact sets; 7/10 episodes reached tick 200; three terminated naturally after all-agent death.
E1: Source population curve	**PENDING**	Profile, logging, launcher, and analysis are implemented; the provider preflight and behavioral evaluation have not yet run.
E2: RecruitmentArena-6	**PENDING**	TFP1, arena, recruitment policies, selectors, oracle, and LLM mechanism studies must be implemented and gated.
E3: Alem recruitment	**PENDING**	Blocked on E2 promotion and a correct opt-in, receiver-specific communication route.

Table 1: Status at the report date. “Pending” denotes an unobserved study, not a null or negative result.

The program asks three ordered questions:

1. **Population scaling:** How does the unchanged Source language agent behave as the number of physical agents changes?
2. **Formation mechanism:** Can six embodied agents form useful, task-bound teams using bounded decentralized recruitment records?
3. **Environment transfer:** Does a promoted recruitment mechanism retain useful performance in Alem when ordinary conversation is restricted to current teammates?

Each stage removes a distinct ambiguity. E0 separates infrastructure from model behavior. E1 measures population behavior before routing is altered. E2 separates recruitment from the full game. E3 tests whether the mechanism survives the navigation, survival, and action constraints of Alem. A failure at an earlier gate blocks downstream efficacy claims.

The completed sequence may support claims about population scaling, decentralized team formation, and team-scoped communication. It does not by itself establish strict information isolation, robustness to adversarial or crashed agents, learned communication, long-horizon role alignment, or populations above six.

2 Shared experimental conventions

2.1 Behavioral model and unchanged Source condition

All paid behavioral stages use homogeneous GPT-5.4 nano action agents with reasoning effort `high`. The Source arm retains:

- the `robust_all` agent;
- the `specific_collaborative` prompt;
- normal reasoning and scratchpad memory;
- the existing action parser and legal-action affordances;
- one model call per physical agent per environment tick; and
- ordinary one-tick-delayed peer broadcast.

No bodyless leader, extra planner, or additional planning call is part of this program. Agent calls within a tick run concurrently. Behavioral arms use one episode worker unless a separate provider-load experiment is preregistered.

2.2 Paired designs and terminal episodes

Comparisons use identical environment seeds across arms. A natural all-agent death is a valid terminal outcome, not an API failure. Every report must state both the requested horizon and actual executed ticks. Cost denominators include actual physical agent-ticks so early death cannot appear to improve efficiency.

2.3 Primary outcomes

The three co-primary Alem outcomes are:

1. paper Total achievement percentage;
2. mean per-agent episode return; and
3. raw unique team achievements.

Supporting outcomes include Base and Coordination achievement percentages, deaths, alive-agent steps, action parse rate, task attempts and successes, messages, emitted bytes, delivered bytes, tokens, model latency, and complete tick wall time. Formation studies additionally report normalized task reward, oracle regret, feasible-team rate, formation latency, churn, roster agreement, and truthful-report calibration.

3 E0: short variable-agent compatibility gate

3.1 Method

E0 crossed $N \in \{1, 2, 3, 4, 6\}$ with seeds 13000 and 13001 for ten ticks in `alem/default`, Easy coordination difficulty, and baseline communication. Exactly one indexed client configuration existed per physical agent, but a deterministic test actor replaced inference. Every actor selected `Noop` and emitted

$$\texttt{E0|AGENT=}i\texttt{|TICK=}t\texttt{|STATE=ACTIVE}.$$

The normal evaluator still performed reset, stepping, action canonicalization, one-tick routing, trajectory and state serialization, debug output, terminal accounting, and attempt-ledger writing.

The gate checked:

- client and agent cardinality and every trajectory agent axis;
- Warrior, Forager, Miner specialization cycling;
- every ordered `Give to Agent` j mapping;
- action parse rate equal to one;
- zero model calls, provider requests, and transport errors;
- exact baseline accounting, $M_{\text{emit}} = NT$ and $M_{\text{deliver}} = N(N-1)T$;
- complete canonical artifacts and ten state snapshots per episode; and
- equality of stable reset and saved-state projection hashes.

One seed-13000 state at every population was also encoded as a 768-by-832 H.264 full-world renderer smoke artifact and validated with `ffprobe`.

3.2 Measured result

N	Episodes	Ticks	Observation dim.	Actions	Give maps	Emitted	Delivered	Delivery bytes	Result
1	2	20	9,476	55	0	20	0	0	**PASS**
2	2	20	9,597	55	2	40	40	1,200	**PASS**
3	2	20	9,718	56	6	60	120	3,600	**PASS**
4	2	20	9,839	57	12	80	240	7,200	**PASS**
6	2	20	10,081	59	30	120	600	18,000	**PASS**

Table 2: E0 measured results. Give mappings count ordered giver–recipient pairs, hence $N(N-1)$.

Across ten episodes, E0 executed 100 environment ticks and 320 scripted agent-ticks, emitted 320 messages, delivered 1,000 copies totaling 30,000 bytes, and saved 100 states. All structural and renderer gates passed. Observation length increased by 121 values for each added teammate. Baseline delivery fan-out grew quadratically. These are interface and accounting facts, not behavioral benefits.

3.3 E0 decision

E0 removed the short-run engineering blocker for the supported $N = 1, 2, 3, 4$ screen and for the separately labeled $N = 6$ extension. It did not test task performance, useful coordination, model reasoning, provider concurrency, recruitment, or role coherence.

4 E0.1: 200-requested-tick compatibility extension

4.1 Method and decision rule

E0.1 repeated the same populations, seeds, scripted actor, Easy environment, and baseline topology with 200 requested ticks. It added monotonic timing for the full episode, the complete evaluator tick, the slowest tick, and the concurrent scripted-worker phase.

Two decisions were fixed:

Structural gate.

Natural termination is allowed, but every episode must produce complete artifacts and pass all shape, action, role, communication, replay, and provider-accounting checks.

Strict horizon gate.

Every episode must execute all 200 requested ticks in addition to passing the structural gate.

4.2 Measured gate outcomes

All ten artifact sets were complete and every structural check passed. Seven episodes reached tick 200. Three ended normally when all agents died from mob combat:

- $N = 1$, seed 13000: 114/200 ticks and one death;
- $N = 4$, seed 13001: 182/200 ticks and four deaths; and
- $N = 6$, seed 13001: 192/200 ticks and six deaths.

Consequently, E0.1 is a structural **PASS** and a strict-horizon **FAIL**. The failure is retained rather than being reclassified after observing that it was a clean environment termination.

N	Actual/requested ticks	Mean return	Full horizon	Alive-step %	Deaths/slots	Mean tick (s)	Mean excl. max (s)	Structural	Horizon
1	314/400	-0.4500	1/2	N/A	1/2	0.151301	0.076475	**PASS**	**FAIL**
2	400/400	-0.5250	2/2	95.88	1/4	0.175483	0.124436	**PASS**	**PASS**
3	400/400	-0.0333	2/2	100.00	0/6	0.235863	0.179381	**PASS**	**PASS**
4	382/400	-0.8625	1/2	78.73	7/8	0.296600	0.236695	**PASS**	**FAIL**
6	392/400	-0.6833	1/2	86.48	8/12	0.437702	0.379037	**PASS**	**FAIL**

Table 3: E0.1 structural, terminal, descriptive return, and timing results. Return is mean cumulative reward per physical agent. "Mean excl. max" removes exactly one slowest tick from each episode and is descriptive rather than a separately timed warm benchmark.

The matrix executed 1,888/2,000 requested environment ticks and 6,194/6,400 requested scripted agent-ticks. It recorded 6,194 emitted messages, 19,544 delivered copies, 614,408 delivered bytes, 1,888 saved states, and zero model calls, provider requests, or transport errors. The summed episode wall time was 498.207821 seconds; end-to-end elapsed time was 659.272638 seconds. The step-weighted complete-tick mean was 0.263203 seconds, and the scripted-worker phase totaled only 0.326201 seconds.

Every achievement, normalized task-reward, player-progression, and monster-kill field was zero. Multi-agent coordination and item-give attempt counts were also zero. The nonpositive return therefore reflects net health change under `Noop`, not task accomplishment. The two-seed returns and mortality patterns cannot rank populations.

4.3 E0.1 decision

E0.1 strengthens the engineering evidence that the evaluator and artifact path support variable populations over substantially longer runs. It does not show that an inactive policy survives to a fixed horizon. Behavioral experiments must treat all-agent death as an outcome and normalize exposure by actual agent-ticks.

5 E1 preregistration: Source performance versus population

5.1 Questions and hypotheses

E1 asks how the unchanged Source condition behaves as physical population changes. It is intentionally descriptive because changing N also changes spawn geometry, role balance, mob slots, prompt size, teammate observations, communication fan-out, and some coordination requirements.

H1a: behavioral scaling.
Useful task performance will vary detectably with population, but no monotonic direction is assumed in advance.

H1b: communication scaling.
Under full broadcast, delivered message copies and bytes will increase approximately with $N(N-1)$ conditional on emission rate.

H1c: efficiency tradeoff.
Any performance gain at larger N may be offset by greater token, communication, and complete-tick cost.

5.2 Fixed configuration and funnel

The behavioral model is GPT-5.4 nano at high reasoning with `robust_all`, `specific_collaborative`, scratchpad, action parser, and free peer broadcast. Debriefs and bulk image generation are disabled. Agents within a tick are called concurrently; episodes use one worker.

The executable profile is `source_scaling_200`. It declares six client slots but the launcher overrides only physical population size; surplus slots are never instantiated as agents. Provider reasoning is controlled by `reasoning_effort=high`; the local-model `agent.reasoning` switch remains unset. Population-and-agent-specific prompt-cache keys use one stable traffic shard, so an interrupted resume returns to the same route rather than reassigning a remaining seed to another seed's shard.

Before any episode call, a dedicated cache route makes one bounded GPT-5.4-nano-high compatibility request and requires an exact parse of `<action>Noop</action>`. Its source commit, dependency lock, resolved configuration hash, response ID, provider status, usage, latency, transport attempts, and effective cache key are written durably. The episode campaign then has a nominal ceiling of 6,000 logical calls for E1a and 3,600 for E1b. Hard campaign ceilings are 12,001 successful logical responses including the preflight and 15,000 observed provider attempts. An exclusive run lock, atomic manifests, exact cell-identity checks, append-only attempt ledgers, and child-process cleanup protect paid resume.

Each terminal artifact includes the `alem-dice-performance-v1` block. Base, Coordination, and Total are reward-weighted paper scores on a 0–100 scale. Unique team achievement counts are binary first-unlocks by achievement type; summed per-agent counts may count the same type once for each attaining agent. They are not repeated world-event counts. Separate cumulative counters retain coordination attempts, resolved attempts, successes, transfers, requests, and revives without summing heterogeneous event units.

Stage	Populations	Seeds	Horizon	Purpose
E1a	1, 2, 3, 4	13100–13102	200	Supported-range screen
E1b	6	13100–13102	200	Gated extension
E1c	Every passing N	13200–13209	1,000	Descriptive curve
E1d	3 and 6	13300–13319	10,000	Paper-scale confirmation

Table 4: E1 staged design. E1d reports Easy, Medium, and Hard separately and begins only after prior gates pass and an updated cost estimate is archived. The campaign was authorized to proceed without another approval pause on 23 July 2026.

E1a must pass at a supported population before that population enters E1c. E1b is separately labeled as an extension beyond the documented one-to-four range. $N = 1$ is plotted as a solo reference because it uses the single-agent wrapper.

5.3 Analysis and planned outputs

The headline figure contains three panels:

1. paper Total achievement percentage;
2. mean per-agent episode return; and
3. raw unique team achievements.

Every panel shows raw seed points and 95% bootstrap intervals. Supporting panels show Base and Coordination achievement percentages, deaths, alive-agent-step exposure, action parsing, total and per-agent-tick tokens, delivered bytes, and complete-tick latency.

Planned figure: Performance versus physical-agent count, with Total percentage, mean per-agent return, and raw unique team achievements as the three primary panels.

Pending result. Populate this subsection only from canonical E1 artifacts. Report early terminations and actual agent-tick denominators. Label the result "fixed-world broadcast population scaling," not a pure causal effect of agent count.

6 E2 preregistration: RecruitmentArena-6

6.1 Purpose and information structure

E2 isolates recruitment before altering Alem. RecruitmentArena is a deterministic, pure-Python six-agent environment. Each episode contains two Warrior-, two Forager-, and two Miner-oriented private capability vectors and private task-specific costs. Public task cards contain a stable task ID, three-dimensional demand, required team size two or three, reward, and deadline.

The information boundary is:

- an agent knows its own true capabilities and costs;
- all agents see public task cards and delivered recruitment records;
- other agents' true capabilities and costs remain private;
- published claims are public but need not equal private truth;
- true values are used only for terminal scoring, calibration, and the analysis-only oracle; and
- no oracle value enters an agent prompt or live allocation decision.

Episodes last 12 recruitment rounds. Control records arrive after a one-round delay. Each agent may emit at most one recruitment-control record per round, capped at 256 UTF-8 bytes. Ordinary conversation is disabled before a team locks and then delivered only to members of that task-bound team. An agent may belong to at most one active team. Membership ends on completion, cancellation, or deadline expiry.

6.2 Scenario families

Four paired scenario families isolate different allocation problems:

1. one feasible complementary task;
2. two disjoint tasks;
3. competing tasks requiring a scarce capability; and
4. oversubscription with more candidates than roster slots.

Formation-only episodes complete a task deterministically when the locked roster's true capabilities satisfy its card. This separates formation quality from embodied execution. E2e later adds a five-round execution bridge.

6.3 Decentralized methods and controls

Open volunteer quorum. Agents publish applications. Every peer derives the same first-valid roster from the delayed public ledger. Selected members explicitly accept before the team locks.

Mutual nomination. Agents publish candidate rosters. A team activates only when every listed member has published the same reciprocal roster.

Peer Contract Net. A deterministic task sponsor is selected from ordinary embodied agents by a stable task-ID hash. The sponsor publishes a call for proposals, candidates bid, the sponsor awards a roster from visible bids, and every selected member accepts.

Controls are:

- one preformed all-six communication team;
- balanced fixed $3+3$ teams;
- no team and no ordinary communication;
- deterministic random allocation; and
- an offline oracle ceiling that never enters an agent prompt.

6.4 Versioned formation protocol

The planned TFP1 control plane includes task announcement, application, nomination, call for proposals, bid, award, accept, decline, lock, cancel, complete, and lease-expiry events. The canonical runtime must enforce:

- strict parsing and canonical serialization;
- the 256-byte UTF-8 limit by rejection, never truncation;
- one team per agent;
- reciprocal roster agreement before lock;
- task-bound completion, cancellation, and expiry;
- one-round delivery delay;
- public control versus team-private ordinary routes;
- deterministic transition sequence numbers and replay hashes; and
- zero ordinary cross-team deliveries.

The runtime validates events and routes. It must not choose an environment action or silently repair a failed model record.

6.5 E2a implementation amendment before canonical execution

A provider-free diagnostic matrix exposed three design ambiguities before any canonical E2 result was created. We therefore froze the following clarifications on 23 July 2026.

All task cards are active concurrently. A locked roster remains committed until a common allocation close after twelve acting rounds and a one-round delivery drain. Feasible teams are scored and completed together at that close. This prevents a scarce agent from being released and reused serially, and makes the live allocation comparable to the static `(tasks + idle)^6` oracle.

The weak Miner vector is $(20, 20, 35)$. The provisional value 65 was invalid for the intended scarcity test: with additive team coverage, a nonspecialist's 20 plus 65 already exceeds the mining demand of 80. Only the strong Miner at $(20, 20, 90)$ can now satisfy either scarce card.

Each decentralized protocol is crossed with two bounded task-choice policies:

`local_commit.`

Each agent selects one task from public cards using only its own capability, its private cost, and a stable public tie-break. Mutual Nomination uses only the public role proxy because that protocol publishes no capability bid. Interest pools are disjoint and communication is low.

`public_sweep`.

Every agent submits at most one bounded record per round while sweeping all tasks in stable order. Candidate overlaps are resolved from the delayed public ledger; no private cross-agent state enters the decision. This spends more control bytes to expose a larger candidate set.

The promotion denominator is oracle-allocation coverage: completed tasks divided by the number simultaneously allocable by the exact oracle. Individually feasible-card formation is reported separately; in the scarce family both cards are individually feasible even though at most one can be allocated. We additionally record achieved utility, utility regret only at equal reward, achieved and oracle raw cost, the oracle assignment, candidate-roster overlap, multi-offer rounds, and roster revisions.

The all-six and fixed $3+3$ controls are topology references rather than exact rosters. Their raw reward and communication are retained, but normalized oracle reward and regret are marked not applicable. Fixed groups receive tasks by public order without consulting hidden capability truth, and no group is reused in the simultaneous allocation window.

6.6 Member-selection ablation

Holding Peer Contract Net fixed, E2d compares:

1. `first_valid`: first claimed-feasible roster by arrival round and agent ID;
2. `random_valid`: deterministic seeded sampling from claimed-feasible rosters; and
3. `exact_utility`: exact enumeration of subsets of the required size.

For claimed capability vector q_i, cost c_{it}, and task demand d_t, the exact rule maximizes

$$U(S,t) = \min_{j:d_{tj}>0} \frac{\sum_{i\in S} q_{ij}}{d_{tj}} - 0.25\frac{\sum_{i\in S} c_{it}}{100|S|}.$$

A roster is claimed-feasible only if it has the requested size and coverage of at least one in every demanded dimension. Utility ties resolve by lexicographically sorted agent IDs. The analysis-only oracle enumerates $(\text{tasks} + \text{idle})^6$ assignments using true capabilities and costs.

6.7 E2 stages and hypotheses

Stage	Seeds	Model use	Purpose
E2a	20000–20999	None	Scripted correctness and determinism
E2b	22000–22002	GPT-5.4 nano high	LLM mechanism screen
E2c	22100–22109	GPT-5.4 nano high	Paired efficacy estimate
E2d	22200–22209	Fixed contract net	Selection-rule ablation
E2e	22300–22309	GPT-5.4 nano high	Five-round execution bridge

Table 5: E2 funnel. Only methods passing the mechanism gate enter E2c.

H2a: formation.

At least one decentralized method will form feasible teams before deadline in at least 80% of feasible opportunities.

H2b: structure.

Contract Net will reduce conflicting rosters relative to open volunteer formation, at the cost of additional control latency.

H2c: selection.

Exact utility will reduce oracle regret relative to first-valid and random selection when scarce capabilities create competing feasible rosters.

H2d: communication boundary.
Team-private execution will use fewer delivered ordinary-message bytes than all-six communication while maintaining complementary completion.

Primary E2 outcomes are normalized task reward and oracle regret. Secondary outcomes are formation latency, feasible-team rate, churn, roster agreement, truthful-report calibration, messages, delivered bytes, tokens, and wall time.

Planned figure: Paired normalized reward and oracle regret by recruitment method, followed by formation latency and delivered-byte efficiency.

Pending result. Record E2a as an engineering result before any paid call. For E2b–E2e, preserve raw model output and rejection codes; do not rewrite invalid messages. If no method passes the mechanism gate, stop before E3, version the protocol, and preserve the negative result.

7 E3 preregistration: recruitment in Alem

7.1 Treatment semantics

E3 introduces recruitment only through opt-in topologies. The baseline topology remains behaviorally unchanged.

- An unteamed agent may emit only one valid bounded TFP1 record on the public bulletin.
- Invalid, oversized, or free-form unteamed messages are rejected, logged, and delivered to no one.
- Once a roster locks, ordinary Source-style `<communication>` text is delivered only to current teammates.
- Nonmembers receive no ordinary team-message copy.
- Public control traffic and private ordinary traffic are charged and reported separately.
- Membership changes take effect deterministically after the accepted lock transition and are replayable from the event journal.
- Middleware validates and routes; it does not select an Alem action.

This is a team-scoped *communication* study, not strict information isolation. Ordinary Alem teammate observations remain visible. Environment Requests and targeted `Give` are not filtered by team in the initial E3 study.

7.2 Observable tasks and stable identity

An embodied agent must locally observe an opportunity before announcing it. Stable task IDs derive only from observable task type, dungeon level, and location. The current text wrapper is insufficient by itself because it collapses multiple same-type coordination cues and normally omits absolute coordinates. E3 therefore requires structured, locally visible opportunity metadata generated from the same light and view masks as the agent's text. This metadata may validate and identify an agent-selected announcement; it must not trigger an automatic announcement or expose a hidden opportunity.

Before E3a, the task taxonomy and capability mapping must be frozen for coordinated mining, construction, elite combat, and handover opportunities. All arms in the first E3 study use the 48-by-48 world and task requirements of two or three agents.

7.3 Critical environment control

At six agents, current world generation sometimes assigns the full physical population as the requirement for mining, construction, and elite-mob coordination. Although an environment field named `max_agents_required` exists, current generation does not apply it. Changing this globally would silently alter the Source scaling environment.

The E3 implementation must therefore add an explicit, experiment-only two-to-three-agent requirement control. It must be enabled identically for every E3 comparison arm and disabled by default so E0/E1 world generation is unchanged.

7.4 Arms, stages, and hypotheses

E3 compares:

1. unchanged six-agent Source full broadcast;
2. balanced fixed $3 + 3$ communication teams;
3. dynamic recruitment using the promoted E2 method and selection rule;
4. a bulletin-only, no-ordinary-conversation screening floor.

The final paper-scale comparison retains Source, fixed $3 + 3$, and dynamic recruitment. A second recruitment method is retained only if its normalized reward is within 5% of the best method while materially reducing communication.

Stage	Environment	Seeds	Horizon	Purpose
E3a	Alem Debug, scripted god mode	14000–14001	20	Route correctness
E3b	Alem Debug, Easy	14100–14102	200	Paid mechanism screen
E3c	Full Alem, Easy	14200–14209	1,000	Paired efficacy estimate
E3d	Full Alem, all difficulties	14300–14319	10,000	Paper scale

Table 6: E3 funnel. E3d requires explicit cost authorization.

H3a: transfer.
A mechanism passing E2 will form valid task teams under Alem's observation and survival constraints.

H3b: communication efficiency.
Dynamic recruitment will reduce ordinary delivered bytes relative to all-six Source broadcast.

H3c: performance preservation.
The communication reduction will not reduce any co-primary outcome by more than the preregistered 5% relative margin in the final paired comparison.

H3d: adaptivity.
Dynamic teams will outperform fixed $3 + 3$ teams on scenarios where task locations and capability needs change during the episode.

Planned figure: Performance–communication frontier for Source broadcast, fixed $3 + 3$, dynamic recruitment, and the bulletin-only floor.

Pending result. E3 results may be added only after E3a demonstrates zero unauthorized ordinary deliveries, deterministic membership replay, and unchanged baseline prompts/routes. Report locally observed task provenance and every rejected control record.

8 Promotion rules and statistical reporting

8.1 Mechanism gates

Every implemented recruitment mechanism must satisfy:

- zero unauthorized cross-team ordinary deliveries;
- 100% roster agreement for every activated team;
- at least 95% action and recruitment-transition validity;
- zero unrecovered provider failures in canonical episodes;
- at least 80% of feasible opportunities formed before deadline;
- matching deterministic replay hashes; and
- no baseline prompt or route change under baseline topology.

These are eligibility gates, not outcomes that may be averaged away. A routing violation blocks efficacy interpretation even if reward is high.

8.2 Behavioral promotion

For a ten-seed behavioral promotion, a treatment must have positive paired direction on at least seven of ten seeds for at least two of Total percentage, mean per-agent return, and unique team achievements, with no material regression in the third. Effect sizes and paired uncertainty are reported even when this directional rule fails.

A final communication-efficiency claim requires a paired 95% interval showing no worse than a 5% relative performance margin on all three co-primary outcomes while reducing delivered communication bytes by at least 30%. Communication cost and model-token cost are reported separately.

8.3 Multiple views of cost

Every behavioral table includes:

- total model calls and calls per actual agent-tick;
- input, output, reasoning, cache-read, and cache-write tokens;
- emitted payload bytes and fan-out-weighted delivered bytes;
- control-plane and ordinary-message bytes separately;
- summed model latency and complete tick wall time; and
- requested versus executed ticks and alive-agent-step exposure.

The coordination-efficiency diagnostic is

$$\text{coordination efficiency} = \frac{\text{paired team-performance improvement}}{\text{delivered communication bytes}},$$

reported alongside its numerator and denominator rather than as a standalone ranking.

9 Implementation readiness and risk register

At the time of this report, E2/E3-specific runtime code, tests, launchers, and configs are not present. Existing infrastructure is useful but insufficient: the evaluator supplies parallel calls, one-tick broadcast, route accounting, and canonical artifacts; E0 supplies six-agent compatibility evidence; and the existing DCP1 parser provides a strict-message pattern. DCP1 does not implement task cards, team membership, leases, byte caps, or receiver-specific routing and must not be relabeled as TFP1.

Table 7: Pre-implementation E2/E3 risk register.

Risk	Failure mode	Required mitigation
Baseline route drift	A routing refactor changes Source delivery, order, delay, or accounting.	Keep the existing baseline branch explicit; add golden prompt and $N(N-1)$ route regressions before dynamic routing.
Global task-size change	Applying a three-agent cap globally changes E1's $N = 4/6$ worlds.	Use an opt-in E3-only requirement control, identical across E3 arms and disabled by default.
Unstable task IDs	Text cues omit absolute location or collapse multiple opportunities.	Expose structured locally visible opportunities from the same visibility mask; never inspect or announce hidden tasks.
Byte-limit corruption	Truncation turns one invalid record into a different apparent record.	Measure the complete UTF-8 payload and reject records above 256 bytes.
Hidden allocation	Middleware or oracle chooses a roster or action using private truth.	Use only published claims for live selection; reserve true values for scoring and the offline oracle.
Roster disagreement	Agents act on different membership states after delayed messages.	Require reciprocal acceptance, canonical transition order, per-agent ledger comparison, and deterministic replay.
Cross-team leakage	One sender-indexed broadcast map delivers private text to all agents.	Use receiver-indexed delivery envelopes for opt-in team topologies and assert zero nonmember recipients.
Protocol noncompliance	LLM output is malformed, oversized, stale, or unauthorized.	Log stable rejection codes, deliver nothing, and never silently rewrite a live record.
Prompt confound	Recruitment instructions accidentally alter the baseline condition.	Inject formation context only under explicit recruitment/fixed-team settings; snapshot baseline prompts.
Premature efficacy claim	Reward is analyzed despite mechanism or replay failure.	Apply structural promotion gates first and preserve stopped/negative studies.
Cost confound	Different episode-worker concurrency changes provider load by arm.	Use one episode worker, concurrent within-tick agents, randomized sequential blocks, and complete usage accounting.

10 Audit and provenance requirements

Every canonical study records:

- resolved configuration and exact CLI invocation;
- Git commit, branch, dirty-source manifest, dependency lock/hash, Python version, and platform;
- model snapshot, reasoning effort, cache policy, and nominal call ceiling;
- scenario or environment seeds and paired-arm mapping;
- raw prompts, raw responses, provider response IDs, finish reasons, usage, retries, and transport errors;

- parsed actions and recruitment events with stable validation codes;
- task cards, event sequence, roster transitions, leases, route envelopes, rejected deliveries, and replay hashes;
- terminal metrics and actual agent-tick denominators; and
- the exact CSV/JSON inputs for every table and figure.

Artifacts from failed or partial attempts are archived rather than overwritten. Canonical episodes require terminal status and no API/runtime error. Resumption retries only failed seeds. Paid stages begin with a dry-run call/token ceiling and a clean committed implementation.

10.1 Completed E0 provenance

E0 ran from `2026-07-23T06:24:16.409143Z` to `2026-07-23T06:28:23.047401Z` at commit `587e7176ce3b02296120e9504c1d7d609e9dcf2a`. Its raw root is `outputs/alem_eval/e0_agent_compatibility_v1`. The source-of-truth result SHA-256 is `7b67b56544a00df63b1e07106c9eeffe51eedc9bc62dc7d8d914af734ad8c47d`.

10.2 Completed E0.1 provenance

The complete E0.1 matrix ran from `2026-07-23T06:57:25.963656Z` to `2026-07-23T07:08:25.236294Z` using measured-run commit `5c4c64338b9aae9132f9cc32b6eec3d746964529`. Its raw root is `outputs/alem_eval/e0_1_agent_compatibility_200_v2`. The source-of-truth result SHA-256 is `6d28e885e241b4f58553291d6fa205fb982878f2575f6bdd3a24d711fda9b529`. The earlier diagnostic attempt remains at `outputs/alem_eval/e0_1_agent_compatibility_200_v1`; it stopped after the two $N = 1$ episodes under the original strict validator.

Both source manifests recorded the pre-existing untracked replay `Results/replays/nano_high_source_full_world.mp4`. It was outside the run roots and was not an input to either result.

11 Interpretation rules and limitations

- E0 and E0.1 establish engineering compatibility only. Their `Noop` returns and mortality are not behavioral scaling evidence.
- E1 is a descriptive fixed-world population curve. It cannot isolate agent count from role balance, spawn geometry, prompt growth, mob pressure, or coordination requirements.
- $N = 1$ uses a different solo wrapper, and $N = 6$ is beyond the paper's documented one-to-four range.
- E2's formation-only reward is a mechanism diagnostic, not evidence that a team can navigate or execute in Alem. E2e and E3 address that gap.
- E3 isolates ordinary message routing, not all channels of information or resource transfer.
- High reward cannot validate a method that leaks messages, disagrees on rosters, or uses hidden information.
- A result from three seeds is a mechanism screen, not confirmatory efficacy evidence. Ten- and twenty-seed stages retain raw paired effects and uncertainty.
- E3d and other 10,000-tick paid stages require explicit authorization after a fresh estimate.

12 Planned final synthesis

The final report will answer the research questions in causal order:

1. Does unchanged Source behavior improve, plateau, or degrade as N increases, and at what communication/token cost?
2. Which decentralized recruitment protocol most reliably forms feasible teams under the bounded bulletin?
3. Does mathematically informed selection reduce oracle regret?
4. Does the promoted mechanism preserve performance when execution and survival are restored?

5. Does team-scoped communication reduce delivered bytes without exceeding the 5% performance-regression margin?

> **Pending result.** Replace this box with a claim-by-evidence table after E1–E3. Every final claim must cite a completed stage, paired seed set, uncertainty statement, mechanism-gate result, and cost outcome. Claims unsupported by a completed stage remain explicitly prospective.

13 Current conclusion

The only completed conclusion is engineering. Alem's canonical language-agent evaluation path supports one, two, three, four, and six physical agents under the tested provider-free conditions. E0 passed its short compatibility gate. E0.1 passed every structural gate but failed the literal full-horizon requirement because three inactive-policy episodes ended naturally.

The behavioral and decentralized-coordination questions remain open. The preregistered E1–E3 sequence is designed to prevent a routing, formatting, hidden-information, or population confound from being mistaken for collective intelligence. This report must be updated from canonical artifacts rather than from console summaries as each downstream gate is completed.

A Reproduction commands for completed gates

E0.

```
uv run -extra baselines-llm -python 3.12 \
  python scripts/run_e0_agent_compatibility.py \
  -output outputs/alem_eval/e0_agent_compatibility_v1
```

E0.1.

```
uv run -extra baselines-llm -python 3.12 \
  python scripts/run_e0_agent_compatibility.py \
  -counts 1,2,3,4,6 \
  -seeds 13000,13001 \
  -steps 200 \
  -output outputs/alem_eval/e0_1_agent_compatibility_200_v2
```

B Planned result-table inventory

1. E1 population-by-seed co-primary outcomes and cost metrics.
2. E1 paired bootstrap intervals and population trend contrasts.
3. E2a scenario-by-method mechanism gates.
4. E2b/c raw validity, formation, reward, regret, and cost.
5. E2d roster-selection utility and oracle-regret contrasts.
6. E2e formation-to-execution attrition and team-private delivery audit.
7. E3a route-invariant and task-provenance audit.
8. E3b/c paired Source, fixed-team, and dynamic-team outcomes.
9. E3 performance–communication noninferiority analysis.
10. Final claim-by-evidence and negative-result table.

C Deferred extensions

Strict filtering of teammate observations, Requests, and `Give`; message loss; crash-stop and Byzantine agents; reputation learning; teams larger than three; populations above six; learned routing; and long-horizon context or role-drift studies are outside E0–E3. They require separate protocols rather than post-hoc interpretation of this program.